

CoRoT-2b

R-2171

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120912 · created 2026-08-13T00:36:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456182.7208 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.065
Transit depth (Rp/Rs)^2	4.99 +/- 2.89 [%]
Orbital Inclination (inc)	89.66 +/- 2.07
Airmass coefficient 1 (a1)	0.87 +/- 0.53
Airmass coefficient 2 (a2)	0.04 +/- 0.35
Scatter in the residuals of the lightcurve fit is	60.28 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.0997 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.065 vs 0.1667 (deviation 0.6σ)
Duration fit vs published	0.0997 d vs 0.095 d (deviation 0.49σ)
Mid-time O–C	-0.5 min
Depth SNR	2.4
Residual scatter	60.28 %

Light curve

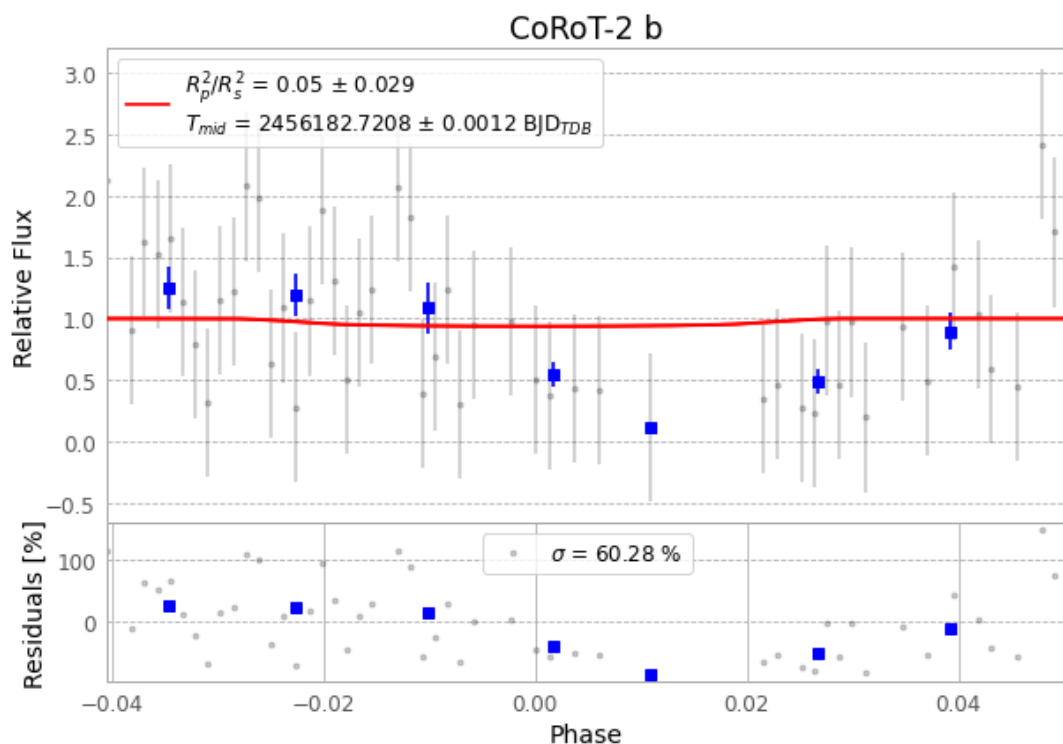


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-09-11.png (SHA-256 243b1472dfe7743c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef20adae7fa63a78...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

78 FITS frames (51 MB), CoRoT-2120912033012.FITS → CoRoT-2120912072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120912.log (14be1388aa1b...), FinalLightCurve_CoRoT-2 b_2012-09-11.png (243b1472dfe7...), FinalLightCurve_CoRoT-2 b_2012-09-11.csv (e1ba806d4083...)

Compute resources

Wall time 174.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:36Z.